MICHIGAN STATE
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29 August 2025

Dear Dr. Mac Gabhann and Dr. Pitzer,

I am pleased to submit the accompanying manuscript "Sonifying character strings into synthetic vocal identity signals and testing identity information detection" for consideration as a research article in PLOS Computational Biology.

Animals can communicate information about their social identities through vocal signals, such as group membership or unique individual identity. Indeed, animals that navigate multiple levels of social organization appear to encode multi-level identity information in their vocalizations. Detecting such complex information encoding could yield new insights into how social and vocal communication systems first originated and continue to change over varying evolutionary timescales.

However, detecting complex information encoding about social identity remains challenging when relying on empirical datasets, as we do not know the true types and amounts of identity information encoded in animals' vocalizations. As a result, it remains very difficult to assess the performance of computational tools when detecting the identity information that animals communicate through vocalizations.

We present a new pipeline to encode identity information across two levels of social structure, the group and individual levels, by sonifying character strings to produce synthetic vocalizations. We encode information in the shape of vocalizations, or frequency modulation patterns. Animals use these acoustic structures to encode both group and individual identity information, as well as to recognize social affiliates. Our pipeline is available as an R package on GitHub for widespread use.

We also performed simulation experiments to test how well spectrographic cross-correlation, a computational tool long applied to bioacoustics datasets, could be used to detect identity information in synthetic vocalizations. In these experiments, we manipulated the complexity of information encoding, as well as the density of the signaling environment, as both factors may influence the distinctiveness of identity information in empirical datasets. Our simulation experiments demonstrate that spectrographic cross-correlation can be used to detect the social level with the most information originally encoded, even under complex information encoding conditions and dense signaling environments.

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Our work introduces a new way of generating synthetic vocalizations through the sonification of character strings. This pipeline produces two representations of vocal identity signals: first as character strings, which can facilitate computationally efficient simulation-based research, as well as audio files, which can be used in empirical experiments with animals. Moreover, we highlight how these synthetic vocalizations can be used to test the performance of a computational tool used widely in ecology and evolution for decades. We anticipate that this work will be of broad interest to researchers who apply computational approaches to address wide-ranging questions about animal behavior in ecology and evolution. Finally, this manuscript could be handled by Associate Editors with expertise in behavioral ecology, language, and learning, including Drs. Chaitanya Gokhale or Barbara Webb.

Sincerely,

Grace Smith-Vidaurre

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